Chapter 19 Homework

Part A multiple choice (select the one that is best in each case. 1 point/question)

1. Which of the following processes is <u>nonspontaneous</u> at 25 °C and 1 atm?
A) formation of glucose from CO_2 and H_2O.
B) dissolution of sugar in a cup of hot coffee.
C) the melting of ice cubes.
D) the neutralization of hydrochloric acid with sodium hydroxide.
E) mixing of water and ethanol.

2. Which one of the following statements is true about the equilibrium constant for a reaction if ΔG^o for the reaction is negative?
A) K = 0 B) K = 1 C) K > 1 D) K < 1 E) More information is needed

3. Predict ΔS_{sys} for the following processes which is positive.
A) Molten gold solidifies.
B) Gaseous Cl_2 dissociates in the stratosphere to form gaseous Cl atoms.
C) Gaseous CO reacts with gaseous H_2 to form liquid methanol, CH_3OH.
D) Calcium phosphate precipitates upon mixing $Ca(NO_3)_2(aq)$ and $(NH_4)_3PO_4(aq)$.
E) Liquid palmitic acid solidifies at 1 °C to solid palmitic acid.

4. Factors that may increase entropy include:________.
(i) the temperature of the system increases, (ii) the volume of a gas increases, (iii) equal volumes of ethanol and water are mixed to form a solution.
A) (i), (ii) and (iii) B) (i) and (iii) C) (ii) and (iii) D) (i) and (ii)

5. The second law of thermodynamics states that ________.
A) $\Delta E = q + w$
B) for any spontaneous process, the entropy of the universe increases
C) the entropy of a pure crystalline substance is zero at absolute zero
D) $\Delta S = q_{rev}/T$ at constant temperature
E) $\Delta H_{rxn}^o = \Sigma\, n\Delta H_f^o$ (products) $- \Sigma\, m\Delta H_f^o$ (reactants)

6. Choose the system with the greater entropy in each case:
A) 1 mol of H_2 (*g*) at 1atm, 298K
B) 1 mol of SO_2 (*g*) at 1atm, 298K
C) 2 mol of NO_2 (*g*) at 1atm, 298K
D) 1 mol of H_2O (*s*) at 1atm, 298K
E) 1 mol of H_2O (*l*) at 1atm, 298K

7. Indicate which of the statement is false.
A) "Translational motion" of molecules refers to their change in spatial location as a function of time.
B) "Rotational" and "vibrational" motions contribute to the entropy in atomic gases like He and Xe.
C) The larger the number of atoms in a molecule, the more degrees of freedom of rotational and vibrational motion it likely has.

D) $CO_2(g)$ and $Ar(g)$ have nearly the same molar mass. At a given temperature, they have different number of microstates.

E) The third law of thermodynamics says that the entropy of a perfect, pure crystal at absolute zero is zero.

8. Describe the reversible process appropriately.

A) A system undergoes a reversible process, the entropy of the universe increases.

B) A system undergoes a reversible process, the change in entropy of the system is exactly matched by an equal and opposite change in the entropy of the surroundings.

C) A system undergoes a reversible process, the entropy change of the system must be zero.

D) Most spontaneous processes in nature are reversible.

9. The following chemical equations describe the same chemical reaction. How do the free energies of these two chemical equations compare?

$$2\ H_2O(l) \rightarrow 2\ H_2(g) + O_2(g) \qquad (1)$$

$$H_2O(l) \rightarrow H_2(g) + 1/2\ O_2(g) \qquad (2)$$

A) $2\Delta G_1^{o} = \Delta G_2^{o}$ B) $\Delta G_1^{o} = \Delta G_2^{o}$ C) $\Delta G_1^{o} = 2\Delta G_2^{o}$ D) None of the above

10. Which of the following statements is true?

A) If $Q = 0$, the system is at equilibrium.

B) The free-energy change for a reaction is independent of temperature.

C) The larger the Q, the larger the ΔG^{o}.

D) If $Q > 1$, $\Delta G > \Delta G^{o}$.

E) If a reaction is spontaneous under standard conditions, it is spontaneous under all conditions.

11. Consider the reaction:

$FeO(s) + Fe(s) + O_2(g) \rightarrow Fe_2O_3(s)$

Given the following table of thermodynamic data at 298 K:

Substance	$\Delta H_f°$(kJ/mol)	$S°$(J/mol-K)
$FeO(s)$	−271.9	60.75
$Fe(s)$	0	27.15
$O_2(g)$	0	205.0
$Fe_2O_3(s)$	−822.16	89.96

The value K for the reaction at 25 °C is ________.

A) 370

B) 5.9×10^{4}

C) 3.8×10^{-14}

D) 7.1×10^{85}

E) 8.1×10^{19}

12. Which of the following correctly defines the second law?

A) In a certain spontaneous process the system undergoes an entropy change of −4.2 J/K; therefore, the entropy change of the surroundings must be larger than 4.2 J/K.

B) The second law of thermodynamics says that entropy is conserved.

C) If the entropy of the system increases during a reversible process, the entropy change of the surroundings must increase by the same amount.

D) The second law of thermodynamics says that the entropy of the system increases for all spontaneous processes.

13. The process of iron being oxidized to make iron (III) oxide (rust) is spontaneous. Which of these statements about this process is/are true?

A) Because the process is spontaneous, the oxidation of iron must be fast.

B) The energy of the universe is decreased when iron is oxidized to rust.

C) Equilibrium is achieved in a closed system when the rate of iron oxidation is equal to the rate of iron (III) oxide reduction.

D) The oxidation of iron is endothermic.

E) The reduction of iron (III) oxide to iron is also spontaneous.

14. The molar heat of vaporization for ethanol is 37.4 kJ/mol. If the boiling point of ethanol is 78.0 °C, calculate $\Delta S°$ in J/K for the vaporization of 0.860 mol ethanol at 78.0 °C.

A) 91.6 J/K B) 0.0916 J/K C) 107 J/K D) 0.106 J/K E) 106 J/K

15. Predict which ΔS of the following process is negative?

A) $N_2(g) + O_2(g) \rightarrow 2\ NO(g)$

B) $4Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(s)$

C) $C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(g)$

D) $H_2O(l) \rightarrow H_2(g) + O_2(g)$

E) $AgCl(s) \rightarrow Ag^+(aq) + Cl^-(aq)$

16. The following reaction and thermodynamic dates are given below

$N_2F_4(g) \rightleftharpoons 2\ NF_2(g)$

$\Delta H° = 85$ kJ; $\Delta S° = 198$ J/K.

The forward reaction is ________.

A) not spontaneous at any temperature.

B) spontaneous at low T but not spontaneous at high T.

C) spontaneous at high T but not spontaneous at low T.

D) spontanous at all temperatures.

17. The thermodynamic quantity that expresses the extent of randomness in a system is ________.

A) entropy

B) heat flow

C) internal energy

D) enthalpy

E) bond energy

18. The K_{sp} for a very insoluble salt is 4.2×10^{-47} at 298 K. What is ΔG° for the dissolution of the salt in water?

A) −2.61 kJ/mol B) −265 kJ/mol C) +115 kJ/mol D) +265 kJ/mol E) −115 kJ/mol

Part B Short question (20 points)

19. Consider a system consisting of an ice cube. **(a)** Under what conditions can the ice cube melt reversibly? **(b)** If the ice cube melts reversibly, is ΔE zero for the process? And give rational reasons.

20. Methanol (CH_3OH) can be made by the controlled oxidation of methane:

$$CH_4(g) + 1/2\ O_2(g) \rightleftharpoons CH_3OH(g)$$

Substance	ΔH_f°(kJ/mol)	S°(J/mol-K)	ΔG°(kJ/mol)
$CH_4(g)$	−74.8	186.3	−50.8
$O_2(g)$	0	205	0
$CH_3OH(g)$	−201.2	237.6	−161.9

(a) Use thermodynamic data to calculate ΔH° and ΔS° for this reaction. **(b)** Predict how ΔG for the reaction varies with increasing temperature, increase, decrease, or stay unchanged. **(c)** Calculate ΔG° at 298 K. Under standard conditions, is the reaction spontaneous at this temperature? **(d)** Is there a temperature at which the reaction would be at equilibrium under standard conditions and that is low enough so that the compounds involved are likely to be stable? Assuming that ΔH° and ΔS° do not change with temperature.

21. Propanol (C_3H_7OH) melts at −126.5 °C and boils at 97.4 °C. Draw a qualitative sketch of how the entropy changes as propanol vapor at 150 °C and 1 atm is cooled to solid propanol at −150 °C and 1 atm.

22. **(a)** Write the chemical equation that defines the normal boiling point of $Br_2(l)$. **(b)** What is the value of ΔG° for the equilibrium in part (a)? **(c)** Use data in the table below to estimate the normal boiling point, for elemental bromine, $Br_2(l)$.

Substance	**ΔH_f° (kJ/mol)**	**S° (J/K·mol)**
$Br_2\ (l)$	0	152.3
$Br_2\ (g)$	30.71	245.3